

Simulation and Modeling

Lecture 15

Input Data Modeling

Fitting to a Theoretical Distribution

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Introduction

- Four steps for modeling input data that fit a theoretical distribution
 - Collection of data from the real systems of interest
 - Requires a substantial time and resource
 - In some cases, data collection might not be possible
 - Limited time
 - Input process might not exist
 - Expert opinion may be used to guess
 - Identification of a probability distribution that represents the input data
 - Development of a frequency distribution, i.e., histogram
 - Choosing the family of distribution from the histogram

Introduction (Contd.)

- Choosing appropriate parameter(s) for the distribution
 - Estimated from the collected data
 - Maximum likelihood estimation
- Evaluation of the chosen distribution and the associated parameters
 - Goodness-of-fit test
 - Informally by graphical method
 - Formally by statistical test
 - Chi-square test
 - Kolmogorov-Smirnov test

Data Collection

- One of the biggest/important tasks
- One of the difficult tasks as well
- Suggestion for successful collection
 - Good planning
 - Analyzing data during collection
 - Collected data are adequate
 - Determining the relationship among input variables
 - Independence of a sequence of data
 - Possibility of auto-correlation
 - Understanding the difference between input and output data
- Independence of collected data
 - Correlation plot
 - Scatter diagram

Correlation plot

- For the following collected data
 - $X_1, X_2, \dots, X_n$ (time order of collection)
 - Correlation plot is a graph of the sample correlation $\hat{\rho}_j, j = 1, 2, \dots, l$
 - Sample correlation $\hat{\rho}_j$ is an estimate of ρ_j between two observation that are j observations apart

$$\hat{\rho}_j = \frac{\hat{c}_j}{s^2(n)} \quad \text{where } \hat{c}_j = \frac{\sum_{i=1}^{n-j} [X_i - \bar{X}(n)][X_{i+j} - \bar{X}(n)]}{n-j}$$

- If $X_1, X_2, \dots, X_n$ are independent,
 - If $\rho_j = 0$, for $j = 1, 2, \dots, n - 1$
 - Though $\hat{\rho}_j$ will not be exactly zero, even when X_i 's are independent since $\hat{\rho}_j$ is an observation of a r.v. with $\mu = 0$
 - If ρ_j 's differ from 0 by a significant amount, then X_i 's are not independent

Correlation plot

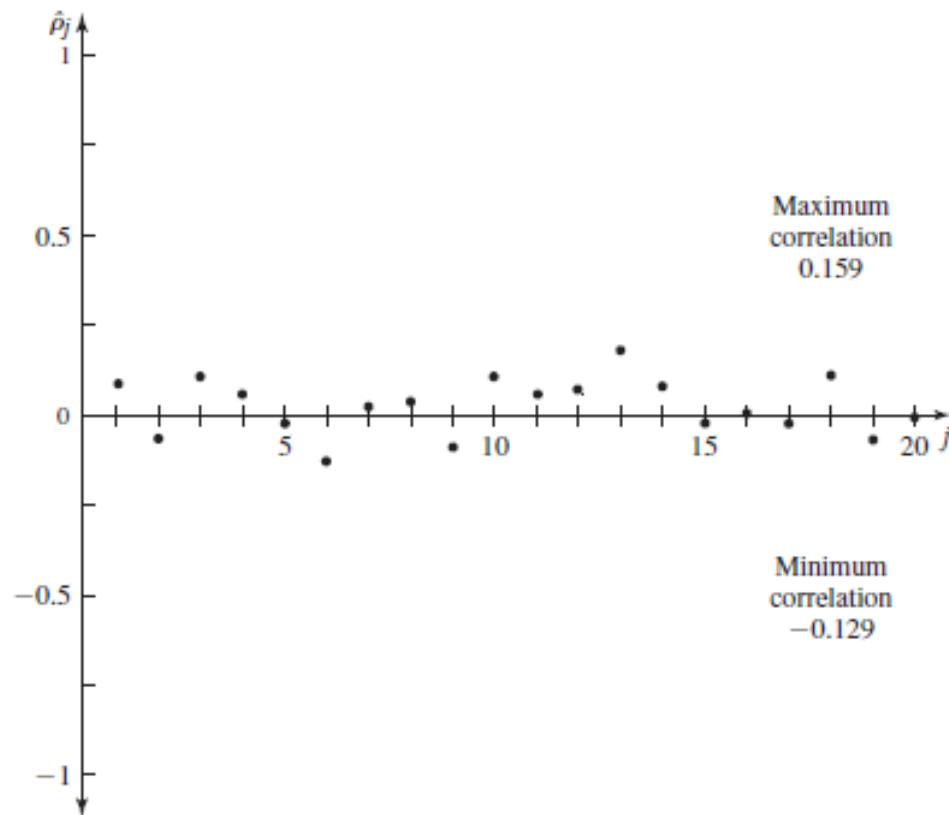


FIGURE 6.27
Correlation plot for independent exponential data.

Scatter Diagram

- Scatter diagram is a plot of the pairs (X_i, X_{i+1}) for $i = 1, 2, \dots, n - 1$
- For non-negative X_i 's, if X_i 's are independent then points (X_i, X_{i+1}) be scattered randomly throughout the first quadrant
- For positive correlation
 - Points will lie along a line with +ve slope
- For negative correlation
 - Points will lie along a line with –ve slope

Scatter Diagram

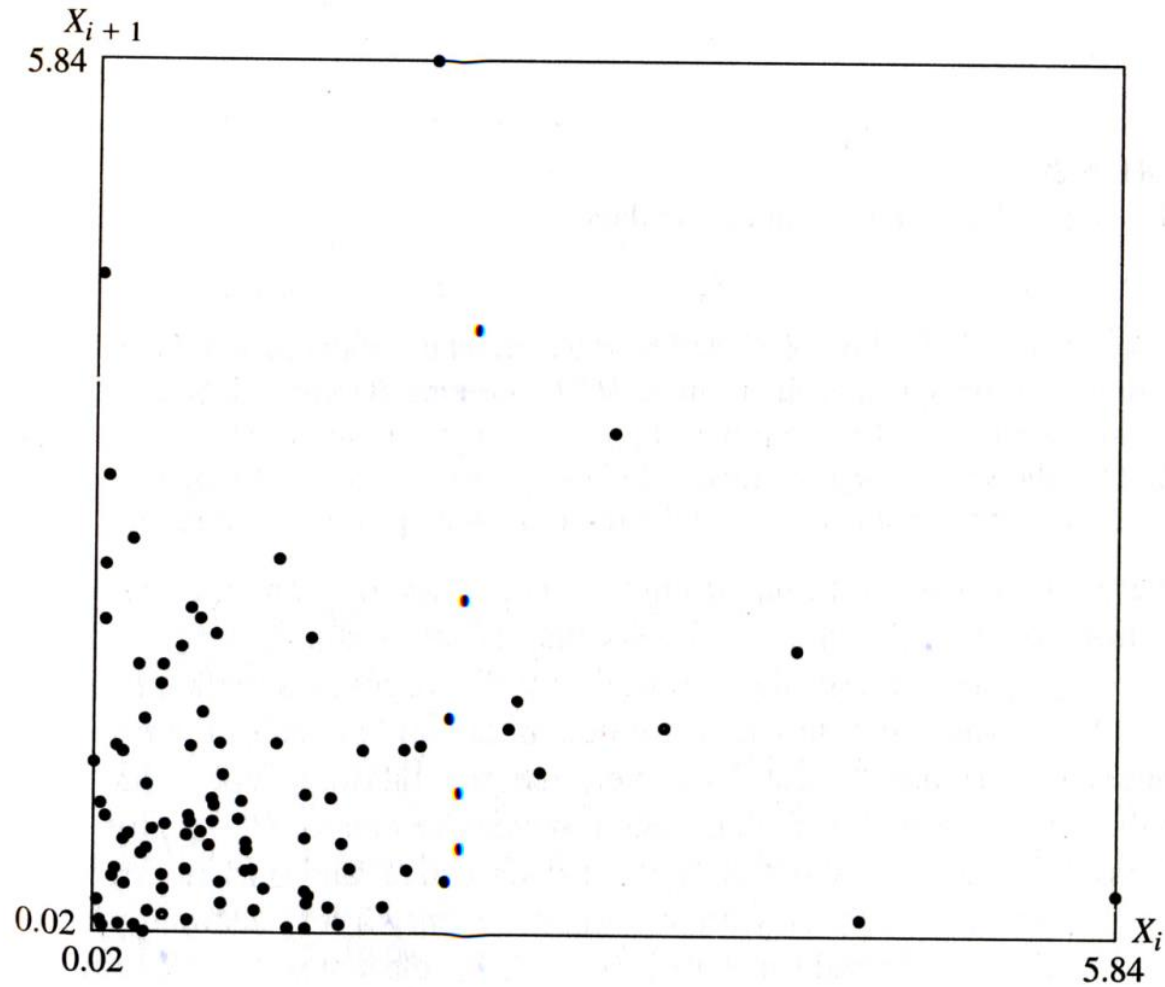


FIGURE 6.28
Scatter diagram for independent exponential data.

Hypothesizing families of distributions

- Summary statistics
- Histogram development
- Choosing families of distribution
- Box plots

Summary Statistics

- Many distributions are partially characterized by functions of their parameters
- Some of these functions are
 - Minimum, maximum, mean, median, variance, coefficient of variation, lexis ratio, skewness
- Values of these functions can be estimated from IID data $X_1, X_2, \dots, X_n$
- Mean and Median
 - For symmetric continuous distribution mean μ is equal to median $x_{0.5}$
 - If $\bar{X}_n$ and $\hat{x}_{0.5}$ almost equal the distribution is symmetric

Summary Statistics – Coefficient of variation

- Coefficient of variation, $cv = \frac{\sqrt{\sigma^2}}{\mu} = \frac{\sqrt{s^2(n)}}{\bar{X}(n)}$
- $cv = 1$, for exponential distribution
- If $\widehat{cv}(n)$ is close to 1, then underlying distribution is suggested as exponential

	$\alpha < 1$	$\alpha = 1$	$\alpha > 1$
Gamma	$cv > 1$	$cv = 1$	$cv < 1$
weibull	$cv > 1$	$cv = 1$	$cv < 1$

- cv is not well defined for distributions like $U(-c, c)$, $c > 0$, or $N(0, \sigma^2)$ since the mean is zero

Summary Statistics – Lexis Ratio (τ)

- Lexis ratio, $\tau = \frac{\sigma^2}{\mu} = \frac{s^2(n)}{\bar{X}(n)}$
- τ is same as cv , only difference is it works for discrete distribution
- Powerful to discriminate among Poisson, Binomial and negative Binomial Distributions

$\tau = 1$	$\tau < 1$	$\tau > 1$
Poisson	Binomial	Negative Binomial
$\tau = \frac{\lambda}{\lambda} = 1$	$\tau = \frac{np(1-p)}{np} = 1-p$	$\tau = \frac{s(1-p)/p^2}{s(1-p)/p} = \frac{1}{p}$

Summary Statistics – Skewness (ν)

- Skewness is a measure of the symmetry of any distributions
- $\nu = \frac{E[(X-\mu)^3]}{(\sigma^2)^{\frac{3}{2}}} = \sum_i^n [X_i - \bar{X}(n)]/S$
- For symmetric distributions, like $N(0, \sigma^2)$, $\nu = 0$
- If $\nu > 0$, distribution has a positive skewness
 - For exponential, $\nu = 2$, skewed to the right
- If $\nu < 0$, distribution has negative skewness

Histogram Development

- Graphical estimation of the density function or mass function
- A histogram is constructed as follows:
 - Divide the range of data into intervals (equal width)
 - Label the horizontal axis to conform to the interval selected
 - Determine the frequency/fraction for each interval
 - Label the vertical axis so that the total occurrence can be plotted
 - Plot the frequency/fraction on the vertical axis

Histogram Development

- Range of values is divided into k intervals
 - $[b_0, b_1), [b_1, b_2), \dots, [b_{k-1}, b_k]$
- Interval width $\Delta b = b_j - b_{j-1}$ is same for all j
- Let h_j be the proportion of X_i 's that are in j -th interval $[b_{j-1}, b_j)$
- Define the function

$$h(x) = \begin{cases} 0, & x < b_0 \\ h_j, & b_{j-1} \leq x \leq b_j, j = 1, 2, \dots, k \\ 0, & x \geq b_k \end{cases}$$

- A histogram is the plot of $h(x)$

Selecting Families of Densities form Histogram

- Usually, by matching the nature of the histogram to the theoretical density functions, families of densities are hypothesized
- h resembles the true density f of the data, why?
- Let X is a RV with density f such that X is distributed as the X_i 's

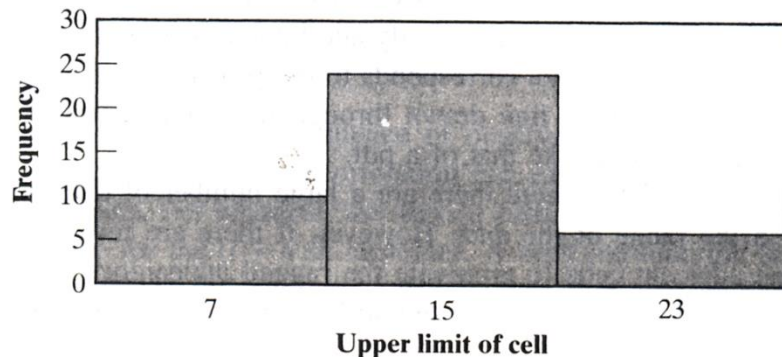
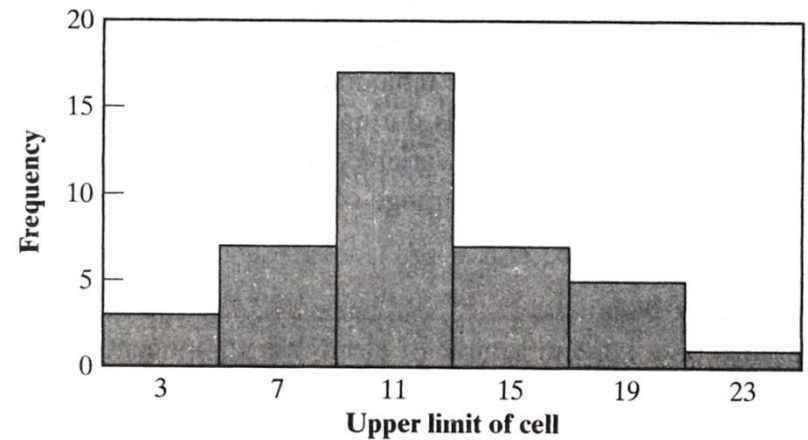
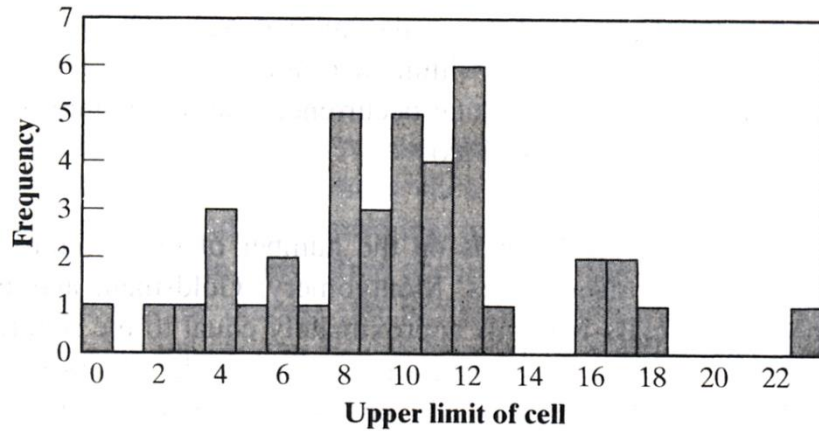
$$p[b_{j-1} \leq X < b_j] = \int_{b_{j-1}}^{b_j} f(x)dx = \Delta b f(x)$$

- The probability that X falls in the j th interval is approximated by
 - $h(y) = h_j \cong \Delta b f(y)$
- $h(y)$ is roughly proportional to $f(y)$
- h and f roughly have the same shape

Histogram – Challenges

- Choosing the number of interval
 - Value of k
- Sturges's rule
 - $k = \lfloor 1 + \log_2 n \rfloor = \lfloor 1 + 3.322 \log_{10} n \rfloor$
- Hines & Montgomery
 - $\hat{k} = \sqrt{n}$
- If k is small, intervals are too wide
 - Histogram will be coarse
- If k is large, intervals are too narrow
 - Histogram will be ragged

Histogram – Example



Histogram - Example

Table 9.1 Number of Arrivals in a 5-Minute Period

<i>Arrivals per Period</i>	<i>Frequency</i>	<i>Arrivals per Period</i>	<i>Frequency</i>
0	12	6	7
1	10	7	5
2	19	8	5
3	17	9	3
4	10	10	3
5	8	11	1

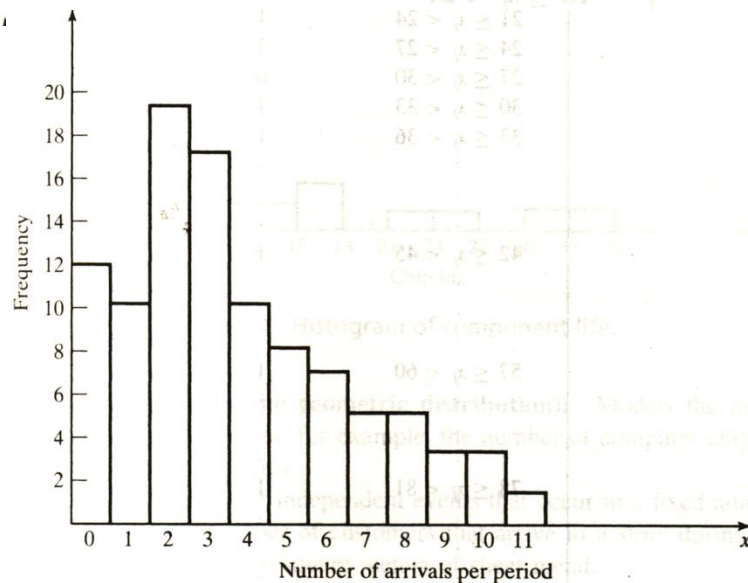


Figure 9.2 Histogram of number of arrivals per period.

Histogram for Composite Distributions

- Histogram will have more than one local modes
- No theoretical distributions provide adequate representation
- For example, consider the machine repair time
 - Two types of breakdown
 - A small portion of repairs are – Major
 - Requires longer time
 - Spare parts are usually ordered for them
 - Most of the repairs are – Minor
 - Requires a shorter time
 - Let p_j is the probability of breakdown type $j = 1, 2$
 - $f(x) = p_1 f_1(x) + p_2 f_2(x)$
 - Composite method can be used to generate random variates

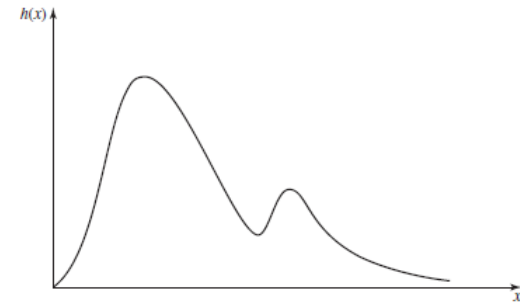


FIGURE 6.31
Histogram corresponding to a density function with two local modes.

Quantile Summary

- Useful to determine whether the density function is
 - Symmetric
 - Skewed to the right
 - Skewed to left

Quantile Summary

- Suppose $F(x)$ is the CDF of a continuous rv
- $F(x)$ is continuous and strictly increasing
 - If $x_1 < x_2$ and $0 < F(x_1) \leq F(x_2) < 1$
 - Then $F(x_1) < F(x_2)$
- For $0 < q < 1$, the q -quantile of $F(x)$ is the number x_q
 - That satisfies, $F(x_q) = q$
- If F^{-1} denote the inverse of $F(x)$ then $x_q = F^{-1}(q)$
- Important q -quantile
 - The median $x_{0.5}$
 - The lower and upper quartiles
 - $x_{0.25}, x_{0.75}$
 - The lower and upper octiles
 - $x_{0.125}, x_{0.875}$

Quantile Summary

TABLE 6.6

Structure of the quantile summary for the sample $X_1, X_2, \dots, X_n$

Quantile	Depth	Sample value(s)		Midpoint
Median	$i = (n + 1)/2$	$X_{(i)}$		$X_{(i)}$
Quartiles	$j = (\lfloor i \rfloor + 1)/2$	$X_{(j)}$	$X_{(n-j+1)}$	$[X_{(j)} + X_{(n-j+1)}]/2$
Octiles	$k = (\lfloor j \rfloor + 1)/2$	$X_{(k)}$	$X_{(n-k+1)}$	$[X_{(k)} + X_{(n-k+1)}]/2$
Extremes	1	$X_{(1)}$	$X_{(n)}$	$[X_{(1)} + X_{(n)}]/2$

- $x_{(i)}$ is an estimate of median
- $x_{(j)}$ and $x_{(n-j+1)}$ are estimates of quartiles
- $x_{(k)}$ and $x_{(n-k+1)}$ are estimates of octiles

Quantile Summary

- If the distribution of X_i 's are symmetric
 - Then the four mid points should be approximately equal
- If the distribution of X_i 's are skewed to the right
 - Then the four mid points should be increasing
- If the distribution of X_i 's are skewed to the left
 - Then the four mid points should be decreasing

TABLE 6.6

Structure of the quantile summary for the sample $X_1, X_2, \dots, X_n$

Quantile	Depth	Sample value(s)		Midpoint
Median	$i = (n + 1)/2$	$X_{(i)}$		$X_{(i)}$
Quartiles	$j = ([i] + 1)/2$	$X_{(j)}$	$X_{(n-j+1)}$	$[X_{(j)} + X_{(n-j+1)}]/2$
Octiles	$k = ([j] + 1)/2$	$X_{(k)}$	$X_{(n-k+1)}$	$[X_{(k)} + X_{(n-k+1)}]/2$
Extremes	1	$X_{(1)}$	$X_{(n)}$	$[X_{(1)} + X_{(n)}]/2$

Quantile	Depth	Sample value(s)		Midpoint
Median	110	0.270		0.270
Quartiles	55.5	0.100	0.545	0.323
Octiles	28	0.050	0.870	0.460
Extremes	1	0.010	1.960	0.985

q-q Plot

- Let $\{X_i, i = 1, 2, \dots, n\}$ are the sample data
- Order the data in increasing order
- Let $\{Y_i, i = 1, 2, \dots, n\}$ be the ordered data
 - $Y_1 \leq Y_2 \leq \dots \leq Y_n$
 - j denotes rank or order number
 - $j = 1$ indicates the smallest data, $j = n$ indicates the largest data
- The q-q plot is based on the fact that Y_i is an estimate of $F^{-1} \left(\frac{j - \frac{1}{2}}{n} \right)$

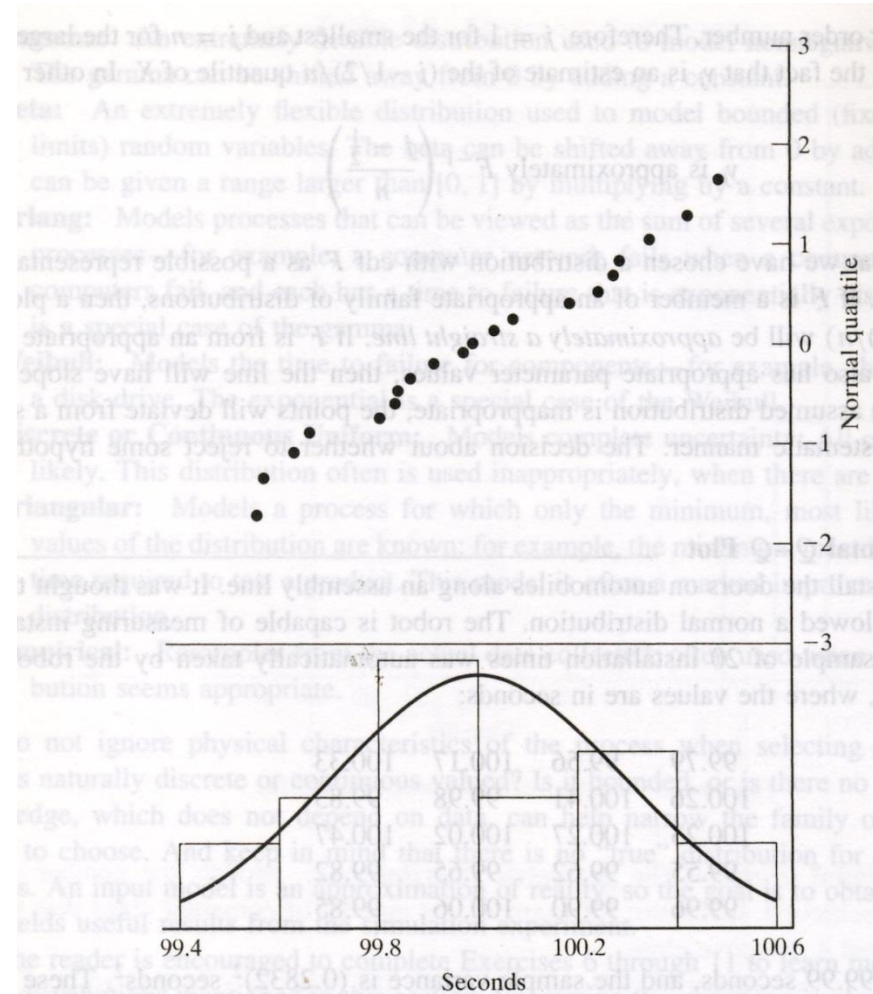
q-q Plot

- Suppose F is the CDF of the chosen distribution
 - Then if F is appropriate then
 - A plot of Y_i vs $F^{-1}\left(\frac{j-\frac{1}{2}}{n}\right)$ is approximately a straight line
 - If F is appropriate and parameter values are appropriate
 - The slope of the line is 1
 - For inappropriate distributions, points deviate from the straight line

q-q Plot

99.79	99.56	100.17	100.33
100.26	100.41	99.98	99.83
100.23	100.27	100.02	100.47
99.55	99.62	99.65	99.82
99.96	99.90	100.06	99.85

<i>j</i>	Value	<i>j</i>	Value	<i>j</i>	Value	<i>j</i>	Value
1	99.55	6	99.82	11	99.98	16	100.26
2	99.56	7	99.83	12	100.02	17	100.27
3	99.62	8	99.85	13	100.06	18	100.33
4	99.65	9	99.90	14	100.17	19	100.41
5	99.79	10	99.96	15	100.23	20	100.47



Parameter Estimation

- $X_1, X_2, \dots, X_n$ random sample from distribution F_θ
 - θ is a set of unknown parameters
 - e.g., when F_θ is Poisson distributed with mean λ
 - $\theta = \{\lambda\}$
 - When F_θ is normal distributed with mean μ and variance σ^2
 - $\theta = \{\mu, \sigma^2\}$
- In statistical inference,
 - A set of value is available as observed data
 - Inference is made about the unknown parameters

Maximum Likelihood Estimators

- Estimator: any statistic used to estimate the value of an (or a set of) unknown parameter θ
- Estimate: Observed value of the estimator is called the estimate
- MLE: Estimator that maximizes the probability of getting the observed data

MLE of a Bernoulli Parameter

- n independent trials, each with success probability p
- What is the maximum likelihood estimator of p ?
- Observed data: $X_1, X_2, \dots, X_n$ where

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{otherwise} \end{cases}$$

- $P[X_i = 1] = p = 1 - P[X_i = 0]$
- $P[X_i = x] = p^x(1-p)^{1-x}, \quad x = 0, 1$

MLE of a Bernoulli Parameter (Contd.)

- The likelihood of observing the data, i.e., the joint probability mass function of the data is
 $P[x_1, x_2, \dots, x_n] = P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n]$

$$= P[X_1 = x_1] \dots P[X_n = x_n]$$

$$= p^{x_1} (1 - p)^{1-x_1} \dots p^{x_n} (1 - p)^{1-x_n}$$

$$= p^{\sum_{i=1}^n x_i} (1 - p)^{n - \sum_{i=1}^n x_i}$$

- The log-likelihood function is

$$\ln P[x_1, \dots, x_n] = \sum_{i=1}^n x_i \ln p + (n - \sum_{i=1}^n x_i) \ln(1 - p)$$